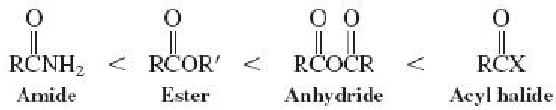
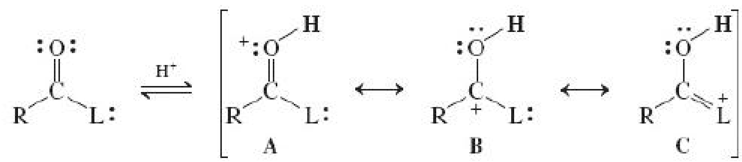
New Reactions

1. Order of Reactivity of Carboxylic Acid Derivatives (Section 20-1)



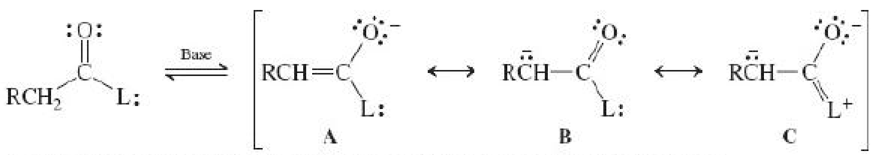
Esters and amides require acid or base catalysts to react with weak nucleophiles.

2. Basicity of the Carbonyl Oxygen (Section 20-1)



L = leaving group
Basicity increases with increasing contribution of resonance structure C.

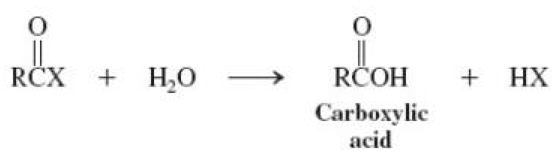
3. Enolate Formation (Section 20-1)



Acidity of the neutral derivative generally increases with decreasing contribution of resonance structure C in the anion.

Reactions of Acyl Halides

4. Water (Section 20-2)



5. Carboxylic Acids (Sections 19-8 and 20-2)

$$R\overset{\displaystyle O}{\overset{\|}{C}}X + R'CO_2H \longrightarrow \underset{\text{Carboxylic anhydride}}{R\overset{\displaystyle O}{\overset{\|}{C}}O\overset{\displaystyle O}{\overset{\|}{C}}R'} + HX$$

6. Alcohols (Section 20-2)

$$R\overset{\displaystyle O}{\overset{\|}{C}}X + R'OH \longrightarrow \underset{\text{Ester}}{R\overset{\displaystyle O}{\overset{\|}{C}}OR'} + \underset{\text{(Removed with pyridine, triethylamine, or other base)}}{HX}$$

7. Amines (Section 20-2)

$$R\overset{\displaystyle O}{\overset{\|}{C}}X + R'NH_2 \longrightarrow \underset{\text{Amide}}{R\overset{\displaystyle O}{\overset{\|}{C}}NHR'} + \underset{\text{(Removed with pyridine, triethylamine, excess } R'NH_2\text{, or other base)}}{HX}$$

8. Cuprate Reagents (Section 20-2)

$$R\overset{\displaystyle O}{\overset{\|}{C}}X \xrightarrow[\;]{\text{1. } R'_2CuLi\text{, THF} \quad \text{2. } H^+\text{, } H_2O} \underset{\text{Ketone}}{R\overset{\displaystyle O}{\overset{\|}{C}}R'} + R'Cu + LiX$$

9. Hydrides (Section 20-2)

$$R\overset{\displaystyle O}{\overset{\|}{C}}X \xrightarrow{\text{1. } LiAl[OC(CH_3)_3]_3H\text{, } (CH_3CH_2)_2O \quad \text{2. } H^+\text{, } H_2O} \underset{\text{Aldehyde}}{R\overset{\displaystyle O}{\overset{\|}{C}}H} + LiX + Al[OC(CH_3)_3]_3$$

Reactions of Carboxylic Acid Anhydrides

10. Water (Section 20-3)

$$R\overset{\displaystyle O}{\overset{\|}{C}}O\overset{\displaystyle O}{\overset{\|}{C}}R + H_2O \longrightarrow \underset{\text{Carboxylic acid}}{2\ R\overset{\displaystyle O}{\overset{\|}{C}}OH}$$

11. Alcohols (Section 20-3)

$$R\overset{O}{\overset{\|}{C}}O\overset{O}{\overset{\|}{C}}R + R'OH \longrightarrow \underset{\textbf{Ester}}{R\overset{O}{\overset{\|}{C}}OR'} + R\overset{O}{\overset{\|}{C}}OH$$

12. Amines ([Section 20-3]())

$$R\overset{O}{\overset{\|}{C}}O\overset{O}{\overset{\|}{C}}R + R'NH_2 \longrightarrow \underset{\textbf{Amide}}{R\overset{O}{\overset{\|}{C}}NHR'} + R\overset{O}{\overset{\|}{C}}OH$$

Reactions of Esters

13. Water (Ester Hydrolysis) ([Sections 19-9]() and [20-4]())

Acid catalysis

$$R\overset{O}{\overset{\|}{C}}OR' + H_2O \xrightarrow{\text{Catalytic } H^+} \underset{\textbf{Carboxylic acid}}{R\overset{O}{\overset{\|}{C}}OH} + R'OH$$

Base catalysis

$$R\overset{O}{\overset{\|}{C}}OR' + \underset{\textbf{1 equivalent}}{{}^{-}OH} \xrightarrow{H_2O} \underset{\textbf{Carboxylate ion}}{R\overset{O}{\overset{\|}{C}}O^-} + R'OH$$

14. Alcohols (Transesterification) and Amines ([Section 20-4]())

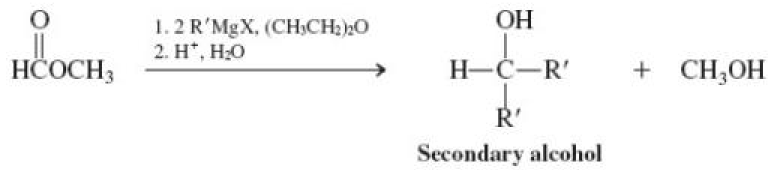
$$R\overset{O}{\overset{\|}{C}}OR' + R''OH \xrightarrow{H^+ \text{ or } {}^{-}OR''} \underset{\textbf{Ester}}{R\overset{O}{\overset{\|}{C}}OR''} + R'OH$$

$$R\overset{O}{\overset{\|}{C}}OR' + R''NH_2 \xrightarrow{\Delta} \underset{\textbf{Amide}}{R\overset{O}{\overset{\|}{C}}NHR''} + R'OH$$

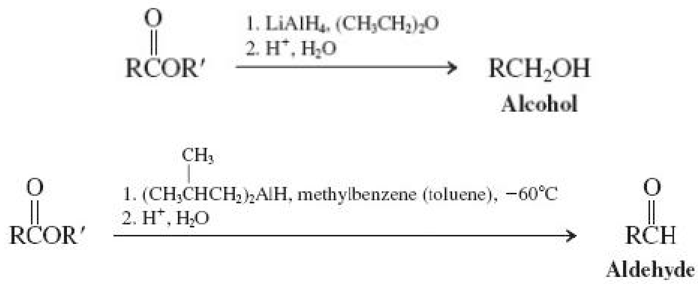
15. Organometallic Reagents ([Section 20-4]())

$$R\overset{O}{\overset{\|}{C}}OR'' \xrightarrow[\text{2. } H^+,\ H_2O]{\text{1. 2 } R'MgX,\ (CH_3CH_2)_2O} \underset{\textbf{Tertiary alcohol}}{R-\overset{OH}{\overset{|}{\underset{R'}{\underset{|}{C}}}}-R'} + R''OH$$

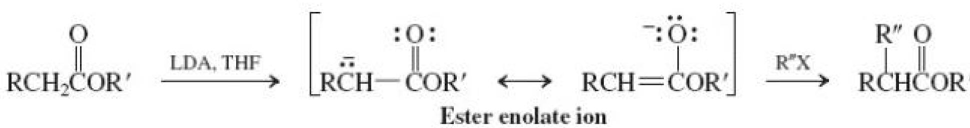
Methyl formate



16. Hydrides (Section 20-4)

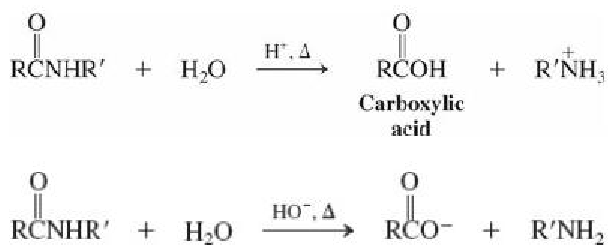


17. Enolates (Section 20-4)

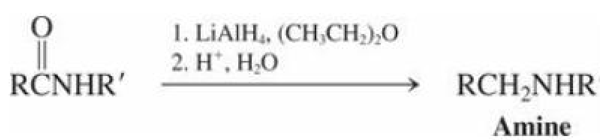


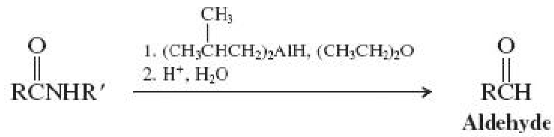
Reactions of Amides

18. Water (Section 20-6)

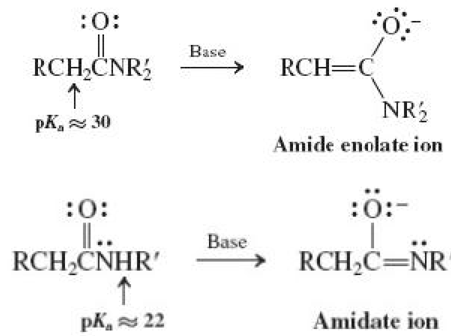


19. Hydrides (Section 20-6)

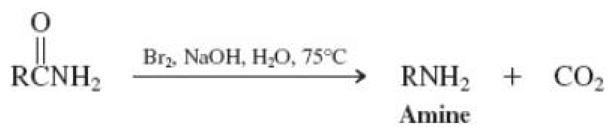




20. Enolates and Amidates (Section 20-7)

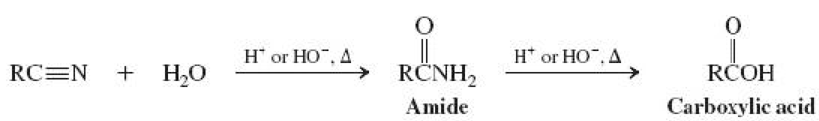


21. Hofmann Rearrangement (Section 20-7)

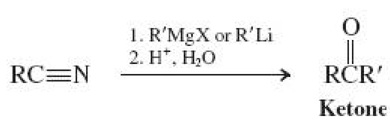


Reactions of Nitriles

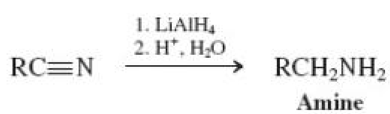
22. Water (Section 20-8)



23. Organometallic Reagents (Section 20-8)



24. Hydrides (Section 20-8)



$$RC{\equiv}N \xrightarrow[\text{2. } H^+,\ H_2O]{\text{1. } (CH_3CH(CH_3)CH_2)_2AlH} \underset{\textbf{Aldehyde}}{RCH{=}O}$$

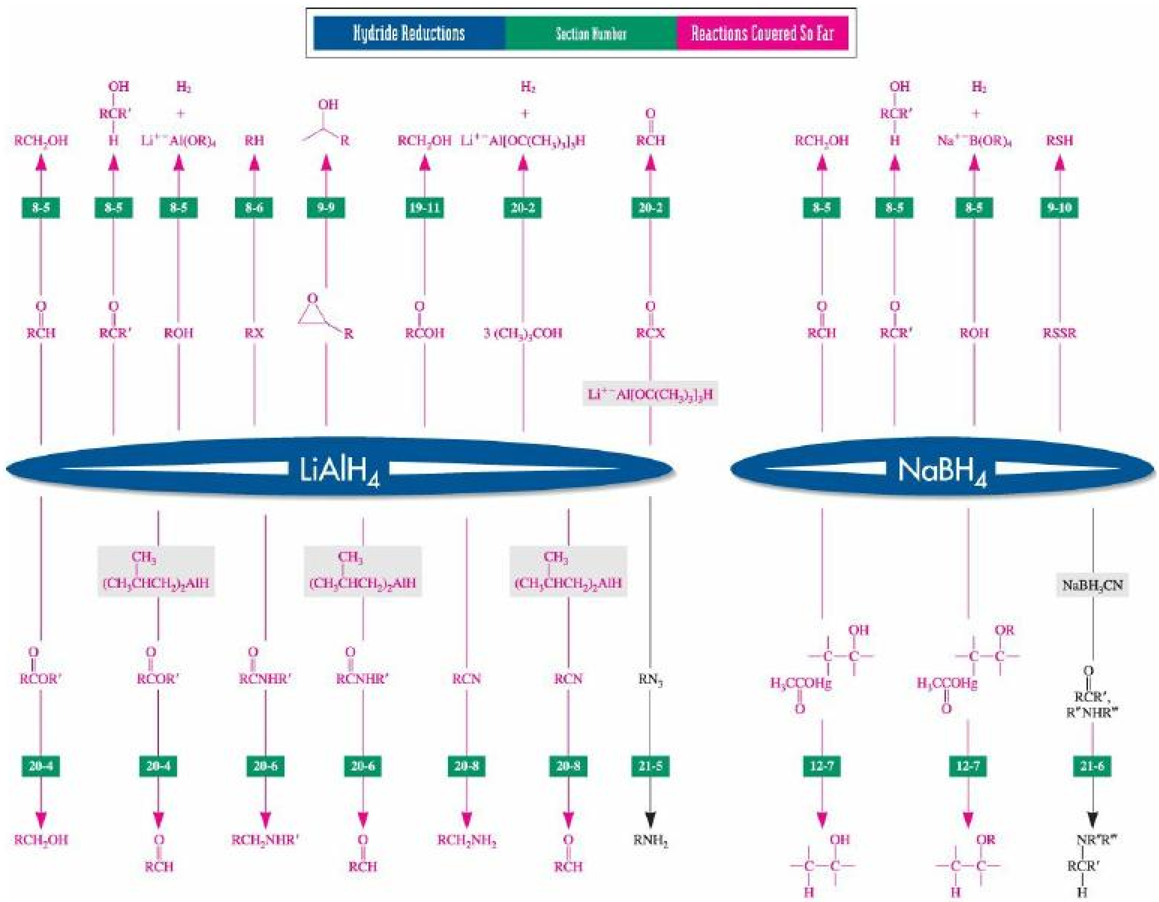
25. Catalytic Hydrogenation ([Section 20-8](#))

$$RC{\equiv}N \xrightarrow{H_2,\ PtO_2,\ CH_3CH_2OH} \underset{\textbf{Amine}}{RCH_2NH_2}$$

Important Concepts

1. The **electrophilic reactivity** of the carbonyl carbon in **carboxylic acid derivatives** is weakened by good electron-donating substituents. This effect, measurable by IR spectroscopy, is responsible not only for the decrease in the reactivity with nucleophiles and acid, but also for the increased basicity along the series: acyl halides-anhydrides-esters-amides. Electron donation by resonance from the nitrogen in amides is so pronounced that there is **hindered rotation** about the amide bond on the NMR time scale.
2. **Carboxylic acid derivatives** are named as **acyl halides, carboxylic anhydrides, alkyl alkanoates, alkanamides,** and **alkanenitriles,** depending on the functional group.
3. **Carbonyl stretching frequencies** in the IR spectra are diagnostic of the carboxylic acid derivatives: Acyl chlorides absorb at v˜=1790-1815 cm-1, $\tilde{v} = 1790 - 1815\ \mathrm{cm}^{-1}$, anhydrides at v˜=1740-1790 and 1800-1850 cm-1, $\tilde{v} = 1740 - 1790 \text{ and } 1800 - 1850\ \mathrm{cm}^{-1}$, esters at v˜=1735-1750 cm-1, $\tilde{v} = 1735 - 1750\ \mathrm{cm}^{-1}$, and amides at v˜=1650-1690 cm-1. $\tilde{v} = 1650 - 1690\ \mathrm{cm}^{-1}$.
4. Carboxylic acid derivatives generally react with **water** (under acid or base catalysis) to hydrolyze to the corresponding carboxylic acid; they combine with **alcohols** to give esters and with **amines** to furnish amides. With **Grignard** and other **organometallic reagents,** they form ketones; esters may react further to form the corresponding alcohols. Reduction by **hydrides** gives products in various oxidation states: aldehydes, alcohols, or amines.
5. Long-chain esters are the constituents of animal and plant **waxes.** Triesters of glycerol are contained in natural **oils** and **fats.** Their hydrolysis gives **soaps. Triglycerides** containing phosphoric acid ester subunits belong to the class of **phospholipids.** Because they carry a highly polar head group and hydrophobic tails, phospholipids form **micelles** and **lipid bilayers.**
6. **Transesterification** can be used to convert one ester into another.

7. The functional group of **nitriles** is somewhat similar to that of the alkynes. The two component atoms are *sp* hybridized. The IR stretching vibration appears at about v˜=2250 cm-1. $\tilde{v} = 2250\ \mathrm{cm}^{-1}$. The hydrogens next to the cyano group are deshielded in ^{1}H NMR. The 13C NMR $^{13}\mathrm{C\ NMR}$ absorptions for nitrile carbons are at relatively low field (δ~112-126 ppm), $(\delta \sim 112–126\ \mathrm{ppm})$, a consequence of the electronegativity of nitrogen.



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